### 3.6.1 Pedagogy for Mathematics

Traditional approaches to teaching mathematics directly jump into abstract symbolic manipulation. This is not very effective in making mathematics accessible to learners. There are several steps before the learner is ready for symbolic manipulations. The first step is to have concrete experiences that embody the mathematical concept involved. Once the learners have immersed themselves in this experience, discussing this experience using language is the next level of abstraction. This language use can then be represented as pictures or diagrams. Finally, these pictures can be converted into the symbols that are used in Mathematics to represent that particular concept or idea. Effective Mathematics pedagogy should take into consideration this sequence for developing a conceptual understanding of mathematics.

For students, problem-solving and problem-posing are critical steps in learning Mathematics. Practice and independent problem solving help students process and remember difficult concepts, and this should be encouraged in the classroom as frequently as possible. Students should also be encouraged to solve problems and puzzles in groups, so that they can see different approaches towards solving a problem and have conversations about mathematical concepts, thereby making them more graspable. They should also be encouraged to pose questions and come up with new problems.

Many students from the Preparatory Stage onwards enjoy learning via scientific experiments performed in laboratories. Students get to experience the following stages of scientific discovery: observing a phenomenon in nature, setting up an experiment in a lab, performing the experiment and noting down observations, trying to find a pattern, and then finally trying to explain the phenomenon. Unfortunately, current practices in Mathematics teaching does not expose students to such a journey. Mathematics is too often presented as a finished product which is purely demonstrative and formal. Guessing using increasingly developed intuition, a skill needed to discover new theorems and their proofs, is discouraged in Mathematics classes.

It is possible to show this 'experimental' nature of Mathematics using an inductive method of teaching at the Preparatory, Middle, and Secondary Stages. The idea is to develop teaching material consisting of appropriate Mathematics experiments. Students can work in groups and guessing can be encouraged. For example, an experiment could students to write down the first few even numbers as sums of two smaller numbers and lead them to observe that they can always be written as sum of 2 primes. Guidance from the Teacher should lead them to conjecture that 'any even number greater than 2 is a sum of 2 primes'. They should collect as much evidence as possible. Though this particular problem is unsolved, towards the end, each student could be given an opportunity to talk about their experiment (instead of proving/ disproving the statement). Many such experiments in number theory, geometry, and combinatorics could be prepared. Such activities have the potential to greatly enhance intuition and problem-solving ability.

Mathematics also naturally provides many opportunities for critical thinking, in the form of interrogating definitions, formulating, or choosing alternative proofs, conjectures, explanations, representations, or generalisations. Curriculum and pedagogy need to provide room and educational opportunity for such thinking. For example, working on match-stick geometry helps interrogate geometric assertions. Students should be encouraged to define their own geometric objects and classes of numbers with specific properties to encourage experimentation, creativity, discovery, and critical thinking.

### 3.6.1.1 Instructional Practices

a. Students should be exposed to multiple ways of seeing the same mathematical concept. This could be through pictures, symbols, different motivations or applications, and different descriptions in spoken language. Each of these provides a distinct perspective, and they together help the student form their own understanding of the mathematical concept.

b. The Teacher should encourage students to express their understanding in their own words using mathematical vocabulary and terms (including in their own home language when different from the medium of instruction). Opportunities should be created for mathematical conversations between students.

c. The Teacher should provide opportunities for students to engage in meaningful discussions involving questions that require explanations (such as 'How could you explain your thinking to someone just beginning to learn this?' or 'How do you know?'). Habits of verification should be inculcated from an early age. When a student has distributed 24 toffees among 8 students, it is important to not only ask 'Has everyone gotten an equal number of toffees?', but also to follow up with 'How do you know?'

d. The Teacher can incorporate problem-solving tasks in the classroom that serve multiple purposes. Problems can be chosen to review concepts previously studied by students and link them to new concepts. The task can be designed so that students have to reason through questions, and then justify their thinking orally or in writing.

e. Teachers can use physical models, diagrams, graphing calculators, simulations, computer algebra systems, games, and other tools to help students model situations, visualise concepts, think through a problem, and devise strategies for solutions.

f. Small-group work can be an effective way of learning Mathematics. Discussions and problem solving in groups give students the opportunity to talk about Mathematics, ask questions they may be hesitant to ask the teacher, and work on harder problems by pooling together their understanding. However, it should be of short duration to manage the groups effectively.

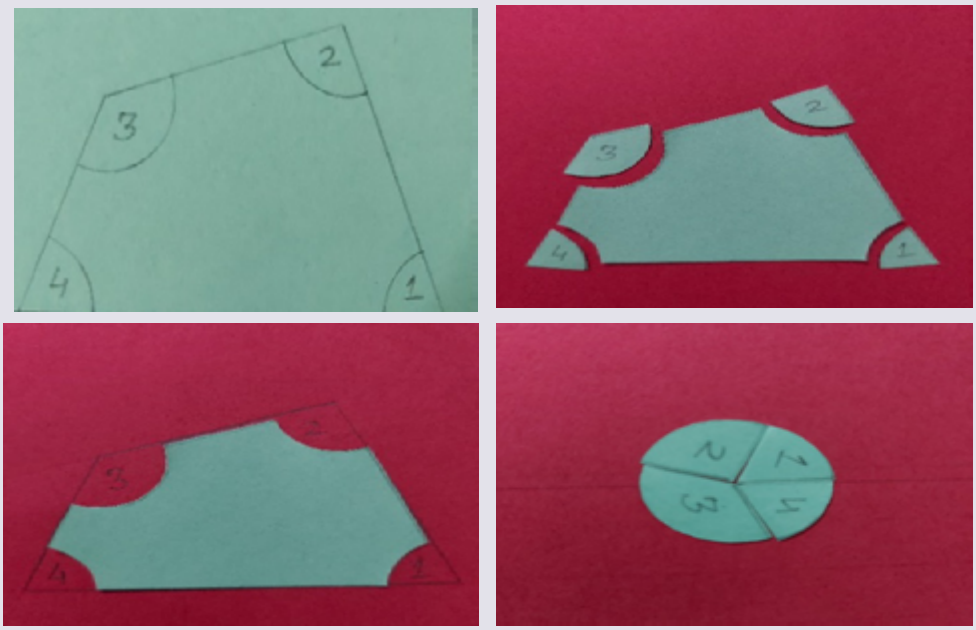
g. Meaningful practice, through worksheets, puzzles, games, mental and oral Mathematics, group work, and homework involving paper and pencil, should be an integral part of the Mathematics classroom. Practice problems should be designed so that students revisit concepts and techniques and see different situations where a certain technique can be used. When choosing problems and guiding students to solve them, teachers should ensure that students are actively learning and not just memorising techniques.

h. During the Middle and Secondary stages, opportunities should be provided for reading simple mathematical text and writing mathematical content. Mathematical communication should be explicitly signalled as an item of teaching and learning. Books and online resource material should be provided by schools for this purpose.

***Teacher's Voice 3.6i***

**Discovering the Magic of Mathematics!**

In geometry, there is a lot of learning through axioms, corollaries, and theorems which often leave students wondering 'Where did this come from!? Does it happen every time!?' So, what I do is I usually introduce such things through activities. For example, to learn that 'sum of all internal angles in a quadrilateral is equal to $360^0$,' we do a simple activity. I ask students to draw any random quadrilateral and then label their internal angles as $\angle 1$, $\angle 2$, $\angle 3$ and $\angle 4$ as shown in the picture below. Next, I ask the students to measure the 4 angles to discover the sum is around $360^0$. Alternatively, when one cuts these angles and join them to meet their all four vertices at a point without leaving any gap (as seen below) to form a complete angle — one discovers that the sum is $360^0$.




Here, my emphasis is always on designing activities that help my students learn mathematical concepts instead of just memorising them as facts and formulae. This is an example of inductive method of teaching theorems, one could also use deductive methods to teach these theorems.

### 3.6.1.2 Some Suggested Methods of Teaching

a. **Play-way (activity-based) method:** In play-way or activity-based learning, students use toys, games, and puzzles to explore mathematics. This may involve physical games, or games/activities involving aids such as dice, puzzles, dominoes, and building blocks Incorporating play can help make some mathematical notions seem more natural and relevant to students, and also enables them to be creative, pose questions, collaborate with others, and use multiple senses to learn. For students who feel alienated by Mathematics, this may help them enjoy the subject and feel more confident in their ability to understand it.

b. **Inquiry-based method:** This method allows students to explore mathematical content by posing, investigating, and answering questions and sharing their findings with their peers for them to critique. Through this method, the student learns to reason and collaborate with their peers to discover mathematical patterns and truths.

c. **Problem-solving method:** Word and logic puzzles (including puzzles that use elimination grids to solve logic problems) are a fun way to teach deductive reasoning. Simple puzzles can help develop students' logical and creative thinking skills in an enjoyable manner.

d. **Inductive method:** Inductive method is based on the principle of induction. Induction is the process to establish a generalised truth by showing that if it is true for a reasonably adequate number of cases, then it is true for all such cases. Thus, the inductive method of

teaching leads us from known to unknown, from a particular case to a general rule and from concrete to abstract. When a number of concrete cases have been understood, the student is able to attempt a generalisation. Students are presented with a series of individual concrete cases, and they are expected to come up with a generalised and abstract mathematical representation of these cases. This method can help students discover patterns in numbers or geometry, which they may later encounter as theorems or formulae. Such discoveries reveal the beauty that drives many people to study mathematics.

e. **Deductive method:** Deduction is the process by which a particular fact is derived from some generally known truths. Thus, in the deductive method of teaching, the student proceeds from general to particular, from abstract to concrete, or from formula to examples. Here, a pre-established rule or formula is given to the student, and they are asked to solve the related problems by using that formula or to prove theorems using definitions, axioms, and postulates.

Each of the methods above has its own advantages and limitations. It is also true that one method does not work for all students, and so the Teacher must draw on their understanding of their class and choose a combination of methods to ensure the learning of every student. The matrix below suggests methods in rows and Stages in three columns.

*Table 3.6i*

| Suggestive Methods | Stages | | |
|---|---|---|---|
| | Preparatory | Middle | Secondary |
| **Play-way** | ✓✓✓ | ✓✓ | ✓ |
| **Discovery/Inquiry** | ✓✓ | ✓✓✓ | ✓✓ |
| **Problem solving** | ✓✓ | ✓✓✓ | ✓✓✓ |
| **Inductive** | ✓✓✓ | ✓✓ | ✓ |
| **Deductive** | ✓ | ✓✓ | ✓✓✓ |
| *Recommendation on Use: ✓✓✓ — More Often, ✓✓ — Often & ✓ — Less Often* | | | |

### 3.6.1.3 Integrating Mathematics with Other Curricular Areas

An interdisciplinary approach enables students to expand their horizons by allowing them to consider and tackle problems that do not fit exactly into one subject. It also changes how students learn by enabling them to synthesise multiple perspectives, instead of driving their thoughts unidirectionally based on the understanding of one discipline. It allows students to explore and involve multiple perspectives and dimensions from different curricular areas to deal with daily life problems. Hence, integrating Mathematics with other Curricular Areas can help students develop interest in the subject and build a holistic view of different disciplines.

Mathematics learning can thus be made more meaningful and interesting by integrating it with other curricular areas. Some possibilities for doing this are described below:

a. **Integrating Mathematics and Art:** Art and Mathematics are closely linked, with both disciplines playing an important role in understanding patterns, as well as enhancing spatial abilities and visualisation. Many activities that are a part of students' lives, such as

music, dance, needlework, and rangoli naturally lead them to see patterns, which can be described and further understood using mathematical language. Integrating the Arts with Mathematics can include art and craft activities that engage students in creating visual patterns, tessellations, and geometric objects, and can include exposure to examples of artworks that contain interesting patterns. Some ideas for integrating Art in the Mathematics classroom are:

i. Creating and analysing different *rangoli/kolam* patterns.

ii. Creating origami, and using it to understand angles, symmetries, and how a 2D object can be transformed into a 3D one.

iii. Recognising geometries and symmetries in art and architecture.

iv. Symmetry can also be explored through dance and movement by assigning mirroring exercises for students. This concept can also be explored through visual games, self-designed board games, simple print-making activities based on traditional art forms like Rogan printing, and by viewing examples of architecture, painting, and sculpture.

v. Pattern activities could also include art forms, like weaving, embroidery, and bead work, where patterning is heavily reliant on mathematical precision, grids, and matrices.

vi. Ratio and proportion are fundamental to the arts. The technique of drawing the human body requires an understanding of proportion (e.g., the length of an arm is about thrice the length of the head). The study of ratios and proportions can also be related to different cultures and their canons of beauty being defined by specific ratios and proportions.

vii. Music is filled with patterns. The joy of making music lies in creating innumerable permutations and combinations of patterns by grouping notes, sounds, and beats. Tempo determines how notes can be combined and fitted into specific rhythm cycles in multiple variations. Music is an extremely useful way to understand fractions since it uses full notes, half notes, quarter notes, and one-eighth notes which also relate to tempo in terms of *ek gun, dugun, trigun, chaugun*. Improvisation in the classical forms of music require an immense alertness and ability to do mental math. For example, creating note patterns in multiples of 3, 5, or 7 in a 4-beat rhythm can be, both challenging and aesthetically pleasing.

   The way frequencies are chosen in music also involves understanding simple fractions, due to what sounds good and most resonant to the ear. For example, the ratio of frequencies of the top and bottom Sa in a *saptak* is 2:1, and the ratio of frequencies of Pa and Sa is 3:2. There are reasons from Physics (namely, the notion of resonance) as to why particular combinations of notes sound good to the ear, and the notes *(shrutis)* that are used in Indian classical music (and also in music around the world), as explained in *Bharata's Natyashastra,* is based on simple whole number ratios of frequencies.

**b. Integrating Mathematics and Sports:** Integrating mathematics and Sports can benefit students who enjoy sports and see the relevance of measurement, unit conversion, probability and statistics, scoring systems, and trajectories of thrown objects in the context of sports. Student projects can explore mathematical connections such as in the Fosbury Flop in high jump or the Duckworth Lewis Scoring System in cricket.

c. **Integrating Mathematics and Science.** The appearance of the Virahanka numbers and the golden angle in nature (e.g., in pinecones, sunflowers, daisies, *kaner* and *tulsi* plants) make for an excellent interdisciplinary journey of discovery.

Similarly, other Curricular Areas can also be integrated with Mathematics to understand and see more meaning of Mathematics in daily life.